



Experiment 1

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Subject Name: Full Stack - II

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1. Aim:

Modern React Foundations – EcoTrack

2. Objective:

- To set up a React project using Vite with proper project structure
- To understand component-based architecture in React
- To apply ES6 array methods (map, filter, reduce) for data-driven UI rendering
- To separate concerns using components, pages, and data modules

3. Implementation/Code:

Header.jsx:

```
EXP1 > exp1 > src > components > JS Header.js > ...
1  const Header = ({ title }) => {
2    return (
3      <header
4        style={{
5          padding: "1rem",
6          backgroundColor: "#27ae60",
7          textAlign: "center",
8        }}
9      >
10     <h1>{title}</h1>
11   </header>
12   );
13 };
14
15 export default Header;
```

Dashboard.jsx:

```
EXP1 > exp1 > src > pages > JS dashboard.js > ...
1  const Dashboard = ({ prop }) => {
2    const Total = prop.reduce((acc, x) => {
3      return acc + x.carbon;
4    }, 0);
5
6    return (
7      <div>
8        <p>Total Carbon Footprint: {Total} Kg</p>
9        <ul>
10         {prop.map((log) => (
11           <li key={log.id}>
12             {log.activity} = {log.carbon} KG
13           </li>
14         ))}
15       </ul>
16     </div>
17   );
18 };
19
20 export default Dashboard;
```

Log.jsx:

```
EXP1 > exp1 > src > pages > JS Logs.js > [?] Logs > [?] filteredArray.map() callback
1  const Logs = ({ logs }) => {
2    const filteredArray = logs.filter((log) => {
3      if (log.carbon >= 4) {
4        return log;
5      }
6    });
7    const Array = logs.filter((log) => {
8      if (log.carbon < 4) {
9        return log;
10      }
11    });
12    return (
13      <ul>
14        <h2>High Carbon Activity</h2>
15        {filteredArray.map((log) => (
16          <li style={{ color: "red" }} key={log.id}>
17            {log.activity} = {log.carbon} Kg
18          </li>
19        ))}
20        <h2>Low Carbon Activity</h2>
21        {Array.map((log) => (
22          <li style={{ color: "green" }} key={log.id}>
23            {log.activity} = {log.carbon} Kg
24          </li>
25        ))}
26      </ul>
27    );
28 };
29
30 export default Logs;
```

App.jsx:

```
p1 > exp1 > src > JS App.js > ...
1 import Dashboard from './pages/dashboard';
2 import logs from './data/logs';
3 import Logs from './pages/Logs';
4
5
6 function App() {
7
8
9   return (
10     <div>
11       <Dashboard prop={logs}/>
12       <Logs logs = {logs}/>
13     </div>
14   )
15 }
16
17 export default App
```

4. Output:

Total Carbon Footprint: 10 Kg

- Car Travel = 4 KG
- Electricity Usage = 6 KG
- Cycling = 0 KG

High Carbon Activity

- Car Travel = 4 Kg
- Electricity Usage = 6 Kg

Low Carbon Activity

- Cycling = 0 Kg

5. Learning Outcomes:

- Analyze Project Structure:

Deduce the purpose and architecture of a React application by examining its file and directory organization.

- Component-Based Architecture:

Understand the distinction between page-level components (pages/) and reusable UI components (components/).

- React Router (or equivalent):

(Assuming App.jsx handles routing) Understand how to implement client-side routing to create a single-page application (SPA) feel.

- Data Handling:

Learn how static data can be imported and utilized within React components (as seen with data/logs.js).